```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings("ignore")

df = pd.read_csv(r"C:\Users\nitin\Desktop\python\project\
Hotel_sales_anyliss\hotel_booking.csv\hotel_booking.csv")
df
```

```
               hotel  is_canceled  lead_time  arrival_date_year  \
0       Resort Hotel            0        342               2015
1       Resort Hotel            0        737               2015
2       Resort Hotel            0          7               2015
3       Resort Hotel            0         13               2015
4       Resort Hotel            0         14               2015
...              ...          ...        ...                ...
119385    City Hotel            0         23               2017
119386    City Hotel            0        102               2017
119387    City Hotel            0         34               2017
119388    City Hotel            0        109               2017
119389    City Hotel            0        205               2017

       arrival_date_month  arrival_date_week_number  \
0                    July                        27
1                    July                        27
2                    July                        27
3                    July                        27
4                    July                        27
...                   ...                       ...
119385             August                        35
119386             August                        35
119387             August                        35
119388             August                        35
119389             August                        35

        arrival_date_day_of_month  stays_in_weekend_nights  \
0                               1                        0
1                               1                        0
2                               1                        0
3                               1                        0
4                               1                        0
...                           ...                      ...
119385                         30                        2
119386                         31                        2
119387                         31                        2
119388                         31                        2
119389                         29                        2

        stays_in_week_nights  adults  ...  customer_type     adr  \
```

```
0                           0       2  ...      Transient    0.00
1                           0       2  ...      Transient    0.00
2                           1       1  ...      Transient   75.00
3                           1       1  ...      Transient   75.00
4                           2       2  ...      Transient   98.00
...                       ...     ...  ...            ...     ...
119385                      5       2  ...      Transient   96.14
119386                      5       3  ...      Transient  225.43
119387                      5       2  ...      Transient  157.71
119388                      5       2  ...      Transient  104.40
119389                      7       2  ...      Transient  151.20

        required_car_parking_spaces total_of_special_requests  \
0                                 0                          0
1                                 0                          0
2                                 0                          0
3                                 0                          0
4                                 0                          1
...                             ...                        ...
119385                            0                          0
119386                            0                          2
119387                            0                          4
119388                            0                          0
119389                            0                          2

        reservation_status reservation_status_date                name
\
0                Check-Out                7/1/2015       Ernest Barnes

1                Check-Out                7/1/2015        Andrea Baker

2                Check-Out                7/2/2015      Rebecca Parker

3                Check-Out                7/2/2015        Laura Murray

4                Check-Out                7/3/2015         Linda Hines

...                    ...                     ...                 ...

119385           Check-Out                9/6/2017     Claudia Johnson

119386           Check-Out                9/7/2017      Wesley Aguilar

119387           Check-Out                9/7/2017        Mary Morales

119388           Check-Out                9/7/2017  Caroline Conley MD

119389           Check-Out                9/7/2017      Ariana Michael


                               email  phone-number       credit_card
```

```
0       Ernest.Barnes31@outlook.com  669-792-1661  ************4322
1            Andrea_Baker94@aol.com  858-637-6955  ************9157
2        Rebecca_Parker@comcast.net  652-885-2745  ************3734
3                 Laura_M@gmail.com  364-656-8427  ************5677
4                LHines@verizon.com  713-226-5883  ************5498
...                             ...           ...               ...
119385          Claudia.J@yahoo.com  403-092-5582  ************8647
119386         WAguilar@xfinity.com  238-763-0612  ************4333
119387     Mary_Morales@hotmail.com  395-518-4100  ************1821
119388      MD_Caroline@comcast.net  531-528-1017  ************7860
119389         Ariana_M@xfinity.com  422-804-6403  ************4482

[119390 rows x 36 columns]
```

```
df.head()
```

```
          hotel  is_canceled  lead_time  arrival_date_year
arrival_date_month  \
0  Resort Hotel            0        342               2015
July
1  Resort Hotel            0        737               2015
July
2  Resort Hotel            0          7               2015
July
3  Resort Hotel            0         13               2015
July
4  Resort Hotel            0         14               2015
July

   arrival_date_week_number  arrival_date_day_of_month  \
0                        27                          1
1                        27                          1
2                        27                          1
3                        27                          1
4                        27                          1

   stays_in_weekend_nights  stays_in_week_nights  adults  ...
customer_type  \
0                        0                     0       2  ...
Transient
1                        0                     0       2  ...
Transient
2                        0                     1       1  ...
Transient
3                        0                     1       1  ...
Transient
4                        0                     2       2  ...
Transient

    adr required_car_parking_spaces total_of_special_requests  \
```

```
0   0.0                              0                          0
1   0.0                              0                          0
2  75.0                              0                          0
3  75.0                              0                          0
4  98.0                              0                          1

  reservation_status reservation_status_date            name  \
0          Check-Out                7/1/2015   Ernest Barnes
1          Check-Out                7/1/2015    Andrea Baker
2          Check-Out                7/2/2015  Rebecca Parker
3          Check-Out                7/2/2015    Laura Murray
4          Check-Out                7/3/2015     Linda Hines

                         email  phone-number       credit_card
0  Ernest.Barnes31@outlook.com  669-792-1661  ************4322
1       Andrea_Baker94@aol.com  858-637-6955  ************9157
2   Rebecca_Parker@comcast.net  652-885-2745  ************3734
3            Laura_M@gmail.com  364-656-8427  ************5677
4           LHines@verizon.com  713-226-5883  ************5498

[5 rows x 36 columns]
```

```
df.shape
```

```
(119390, 36)
```

```
df.columns
```

```
Index(['hotel', 'is_canceled', 'lead_time', 'arrival_date_year',
       'arrival_date_month', 'arrival_date_week_number',
       'arrival_date_day_of_month', 'stays_in_weekend_nights',
       'stays_in_week_nights', 'adults', 'children', 'babies', 'meal',
       'country', 'market_segment', 'distribution_channel',
       'is_repeated_guest', 'previous_cancellations',
       'previous_bookings_not_canceled', 'reserved_room_type',
       'assigned_room_type', 'booking_changes', 'deposit_type',
'agent',
       'company', 'days_in_waiting_list', 'customer_type', 'adr',
       'required_car_parking_spaces', 'total_of_special_requests',
       'reservation_status', 'reservation_status_date', 'name',
'email',
       'phone-number', 'credit_card'],
      dtype='str')
```

```
var = df.drop(columns = ['name', 'email',
       'phone-number', 'credit_card'])
```

```
var.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
```

```
Data columns (total 32 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   hotel                           119390 non-null  str
 1   is_canceled                     119390 non-null  int64
 2   lead_time                       119390 non-null  int64
 3   arrival_date_year               119390 non-null  int64
 4   arrival_date_month              119390 non-null  str
 5   arrival_date_week_number        119390 non-null  int64
 6   arrival_date_day_of_month       119390 non-null  int64
 7   stays_in_weekend_nights         119390 non-null  int64
 8   stays_in_week_nights            119390 non-null  int64
 9   adults                          119390 non-null  int64
 10  children                        119386 non-null  float64
 11  babies                          119390 non-null  int64
 12  meal                            119390 non-null  str
 13  country                         118902 non-null  str
 14  market_segment                  119390 non-null  str
 15  distribution_channel            119390 non-null  str
 16  is_repeated_guest               119390 non-null  int64
 17  previous_cancellations          119390 non-null  int64
 18  previous_bookings_not_canceled  119390 non-null  int64
 19  reserved_room_type              119390 non-null  str
 20  assigned_room_type              119390 non-null  str
 21  booking_changes                 119390 non-null  int64
 22  deposit_type                    119390 non-null  str
 23  agent                           103050 non-null  float64
 24  company                         6797 non-null    float64
 25  days_in_waiting_list            119390 non-null  int64
 26  customer_type                   119390 non-null  str
 27  adr                             119390 non-null  float64
 28  required_car_parking_spaces     119390 non-null  int64
 29  total_of_special_requests       119390 non-null  int64
 30  reservation_status              119390 non-null  str
 31  reservation_status_date         119390 non-null  str
dtypes: float64(4), int64(16), str(12)
memory usage: 29.1 MB
```

```
var["reservation_status_date"]
=pd.to_datetime(var["reservation_status_date"])
```

```
var.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 119390 entries, 0 to 119389
Data columns (total 32 columns):
 #   Column                          Non-Null Count   Dtype
---  ------                          --------------   -----
 0   hotel                           119390 non-null  str
 1   is_canceled                     119390 non-null  int64
```

```
 2   lead_time                       119390 non-null  int64
 3   arrival_date_year               119390 non-null  int64
 4   arrival_date_month              119390 non-null  str
 5   arrival_date_week_number        119390 non-null  int64
 6   arrival_date_day_of_month       119390 non-null  int64
 7   stays_in_weekend_nights         119390 non-null  int64
 8   stays_in_week_nights            119390 non-null  int64
 9   adults                          119390 non-null  int64
 10  children                        119386 non-null  float64
 11  babies                          119390 non-null  int64
 12  meal                            119390 non-null  str
 13  country                         118902 non-null  str
 14  market_segment                  119390 non-null  str
 15  distribution_channel            119390 non-null  str
 16  is_repeated_guest               119390 non-null  int64
 17  previous_cancellations          119390 non-null  int64
 18  previous_bookings_not_canceled  119390 non-null  int64
 19  reserved_room_type              119390 non-null  str
 20  assigned_room_type              119390 non-null  str
 21  booking_changes                 119390 non-null  int64
 22  deposit_type                    119390 non-null  str
 23  agent                           103050 non-null  float64
 24  company                         6797 non-null    float64
 25  days_in_waiting_list            119390 non-null  int64
 26  customer_type                   119390 non-null  str
 27  adr                             119390 non-null  float64
 28  required_car_parking_spaces     119390 non-null  int64
 29  total_of_special_requests       119390 non-null  int64
 30  reservation_status              119390 non-null  str
 31  reservation_status_date         119390 non-null  datetime64[us]
dtypes: datetime64[us](1), float64(4), int64(16), str(11)
memory usage: 29.1 MB
```

```
var.describe(include = "object")
```

```
             hotel arrival_date_month    meal country
market_segment  \
count       118897              118897  118897  118897         118897

unique           2                  12       5     177              7

top     City Hotel              August      BB     PRT      Online TA

freq         79301               13852   91862   48585          56402


       distribution_channel reserved_room_type assigned_room_type  \
count                118897              118897              118897
unique                    5                  10                  12
top                   TA/TO                   A                   A
```

```
freq                   97729               85600               73862

       deposit_type customer_type reservation_status
count        118897        118897             118897
unique            3             4                  3
top      No Deposit     Transient          Check-Out
freq         104163         89173              74745
```

```python
for col in var.describe(include = "object").columns:
    print(col)
    print(var[col].unique())
    print("-"*50)
```

```
hotel
<StringArray>
['Resort Hotel', 'City Hotel']
Length: 2, dtype: str
--------------------------------------------------
arrival_date_month
<StringArray>
[     'July',    'August', 'September',   'October',  'November',
'December',
   'January',  'February',     'March',     'April',       'May',
'June']
Length: 12, dtype: str
--------------------------------------------------
meal
<StringArray>
['BB', 'FB', 'HB', 'SC', 'Undefined']
Length: 5, dtype: str
--------------------------------------------------
country
<StringArray>
['PRT', 'GBR', 'USA', 'ESP', 'IRL', 'FRA', 'ROU', 'NOR', 'OMN', 'ARG',
 ...
 'ATA', 'GTM', 'ASM', 'MRT', 'NCL', 'KIR', 'SDN', 'ATF', 'SLE', 'LAO']
Length: 177, dtype: str
--------------------------------------------------
market_segment
<StringArray>
[       'Direct',     'Corporate',     'Online TA', 'Offline TA/TO',
 'Complementary',        'Groups',      'Aviation']
Length: 7, dtype: str
--------------------------------------------------
distribution_channel
<StringArray>
['Direct', 'Corporate', 'TA/TO', 'Undefined', 'GDS']
Length: 5, dtype: str
--------------------------------------------------
reserved_room_type
```

```
<StringArray>
['C', 'A', 'D', 'E', 'G', 'F', 'H', 'L', 'B', 'P']
Length: 10, dtype: str
--------------------------------------------------
assigned_room_type
<StringArray>
['C', 'A', 'D', 'E', 'G', 'F', 'I', 'B', 'H', 'L', 'K', 'P']
Length: 12, dtype: str
--------------------------------------------------
deposit_type
<StringArray>
['No Deposit', 'Refundable', 'Non Refund']
Length: 3, dtype: str
--------------------------------------------------
customer_type
<StringArray>
['Transient', 'Contract', 'Transient-Party', 'Group']
Length: 4, dtype: str
--------------------------------------------------
reservation_status
<StringArray>
['Check-Out', 'Canceled', 'No-Show']
Length: 3, dtype: str
--------------------------------------------------
```

```
var.isnull().sum()
```

```
hotel                                   0
is_canceled                             0
lead_time                               0
arrival_date_year                       0
arrival_date_month                      0
arrival_date_week_number                0
arrival_date_day_of_month               0
stays_in_weekend_nights                 0
stays_in_week_nights                    0
adults                                  0
children                                4
babies                                  0
meal                                    0
country                               488
market_segment                          0
distribution_channel                    0
is_repeated_guest                       0
previous_cancellations                  0
previous_bookings_not_canceled          0
reserved_room_type                      0
assigned_room_type                      0
booking_changes                         0
deposit_type                            0
```

```
agent                                16340
company                             112593
days_in_waiting_list                     0
customer_type                            0
adr                                      0
required_car_parking_spaces              0
total_of_special_requests                0
reservation_status                       0
reservation_status_date                  0
dtype: int64
```

```
var.drop(["company", "agent"], axis = 1, inplace = True)
```

```
var.dropna(inplace = True)
```

```
var.isnull().sum()
```

```
hotel                               0
is_canceled                         0
lead_time                           0
arrival_date_year                   0
arrival_date_month                  0
arrival_date_week_number            0
arrival_date_day_of_month           0
stays_in_weekend_nights             0
stays_in_week_nights                0
adults                              0
children                            0
babies                              0
meal                                0
country                             0
market_segment                      0
distribution_channel                0
is_repeated_guest                   0
previous_cancellations              0
previous_bookings_not_canceled      0
reserved_room_type                  0
assigned_room_type                  0
booking_changes                     0
deposit_type                        0
days_in_waiting_list                0
customer_type                       0
adr                                 0
required_car_parking_spaces         0
total_of_special_requests           0
reservation_status                  0
reservation_status_date             0
dtype: int64
```

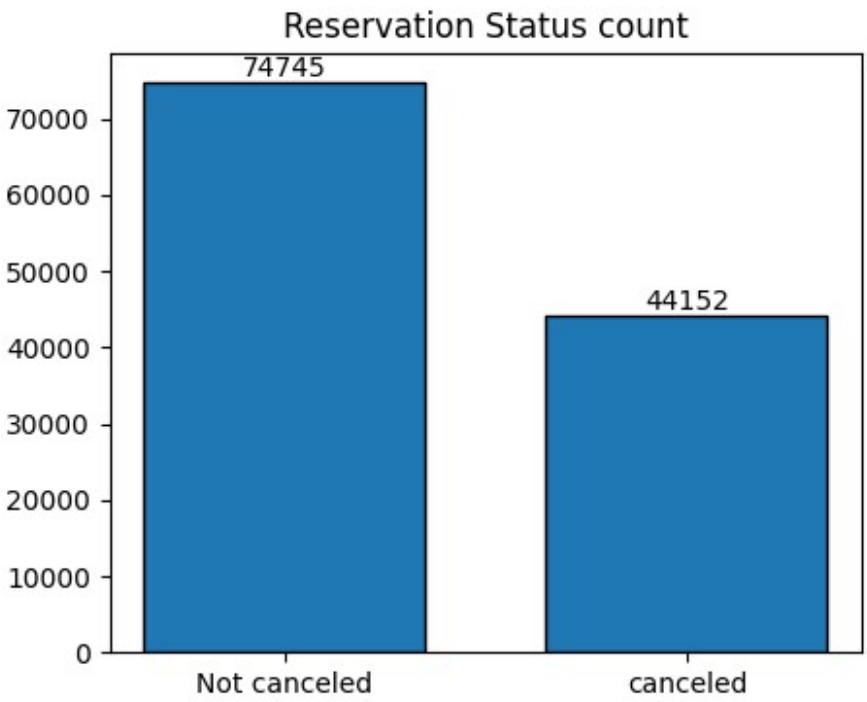
```
var.describe()
```

```
         is_canceled      lead_time  arrival_date_year  \
count  118898.000000  118898.000000      118898.000000
mean        0.371352     104.311435        2016.157656
min         0.000000       0.000000        2015.000000
25%         0.000000      18.000000        2016.000000
50%         0.000000      69.000000        2016.000000
75%         1.000000     161.000000        2017.000000
max         1.000000     737.000000        2017.000000
std         0.483168     106.903309           0.707459

       arrival_date_week_number  arrival_date_day_of_month  \
count             118898.000000              118898.000000
mean                 27.166555                  15.800880
min                   1.000000                   1.000000
25%                  16.000000                   8.000000
50%                  28.000000                  16.000000
75%                  38.000000                  23.000000
max                  53.000000                  31.000000
std                  13.589971                   8.780324

       stays_in_weekend_nights  stays_in_week_nights         adults  \
count            118898.000000         118898.000000  118898.000000
mean                 0.928897              2.502145       1.858391
min                  0.000000              0.000000       0.000000
25%                  0.000000              1.000000       2.000000
50%                  1.000000              2.000000       2.000000
75%                  2.000000              3.000000       2.000000
max                 16.000000             41.000000      55.000000
std                  0.996216              1.900168       0.578576

            children         babies  is_repeated_guest  \
count  118898.000000  118898.000000      118898.000000
mean        0.104207       0.007948           0.032011
min         0.000000       0.000000           0.000000
25%         0.000000       0.000000           0.000000
50%         0.000000       0.000000           0.000000
75%         0.000000       0.000000           0.000000
max        10.000000      10.000000           1.000000
std         0.399172       0.097380           0.176029

       previous_cancellations  previous_bookings_not_canceled  \
count           118898.000000                   118898.000000
mean                 0.087142                        0.131634
min                  0.000000                        0.000000
25%                  0.000000                        0.000000
50%                  0.000000                        0.000000
75%                  0.000000                        0.000000
max                 26.000000                       72.000000
std                  0.845869                        1.484672
```

```
       booking_changes  days_in_waiting_list            adr  \
count    118898.000000         118898.000000  118898.000000
mean          0.221181              2.330754     102.003243
min           0.000000              0.000000      -6.380000
25%           0.000000              0.000000      70.000000
50%           0.000000              0.000000      95.000000
75%           0.000000              0.000000     126.000000
max          21.000000            391.000000    5400.000000
std           0.652785             17.630452      50.485862

       required_car_parking_spaces  total_of_special_requests  \
count                 118898.000000              118898.000000
mean                       0.061885                   0.571683
min                        0.000000                   0.000000
25%                        0.000000                   0.000000
50%                        0.000000                   0.000000
75%                        0.000000                   1.000000
max                        8.000000                   5.000000
std                        0.244172                   0.792678

          reservation_status_date
count                      118898
mean   2016-07-30 07:37:53.336809
min           2014-10-17 00:00:00
25%           2016-02-02 00:00:00
50%           2016-08-08 00:00:00
75%           2017-02-09 00:00:00
max           2017-09-14 00:00:00
std                           NaN

var = var[var["adr"]< 5000]
```

Data Analysis and Visualization

```
cancel_percentage = var["is_canceled"].value_counts(normalize = True)
print(cancel_percentage)

plt.figure(figsize = (5,4))
plt.title("Reservation Status count")
x = plt.bar(["Not canceled", "canceled"],
var["is_canceled"].value_counts(), edgecolor = "black", width = 0.7)
plt.bar_label(x)
plt.show()

is_canceled
0    0.628653
1    0.371347
Name: proportion, dtype: float64
```



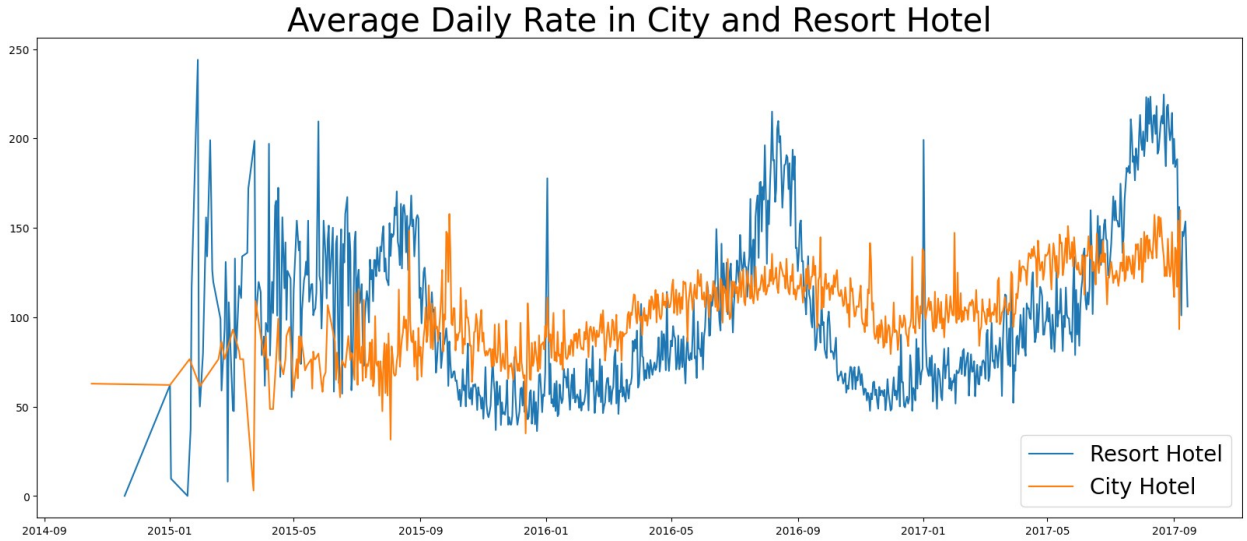
Most bookings are completed, but the number of cancellations is still significant.

```
plt.figure(figsize = (8,4))
ax = sns.countplot(x = 'hotel' , hue = 'is_canceled', data = var,
palette = "inferno")

plt.title("Reservation status in different hotels", size = 20)
plt.xlabel('hotel')
plt.ylabel('number of reservation')

plt.legend(["not canceled","canceled"])
plt.show()
```

City hotels receive more reservations than resort hotels, and they also experience a higher number of cancellations.

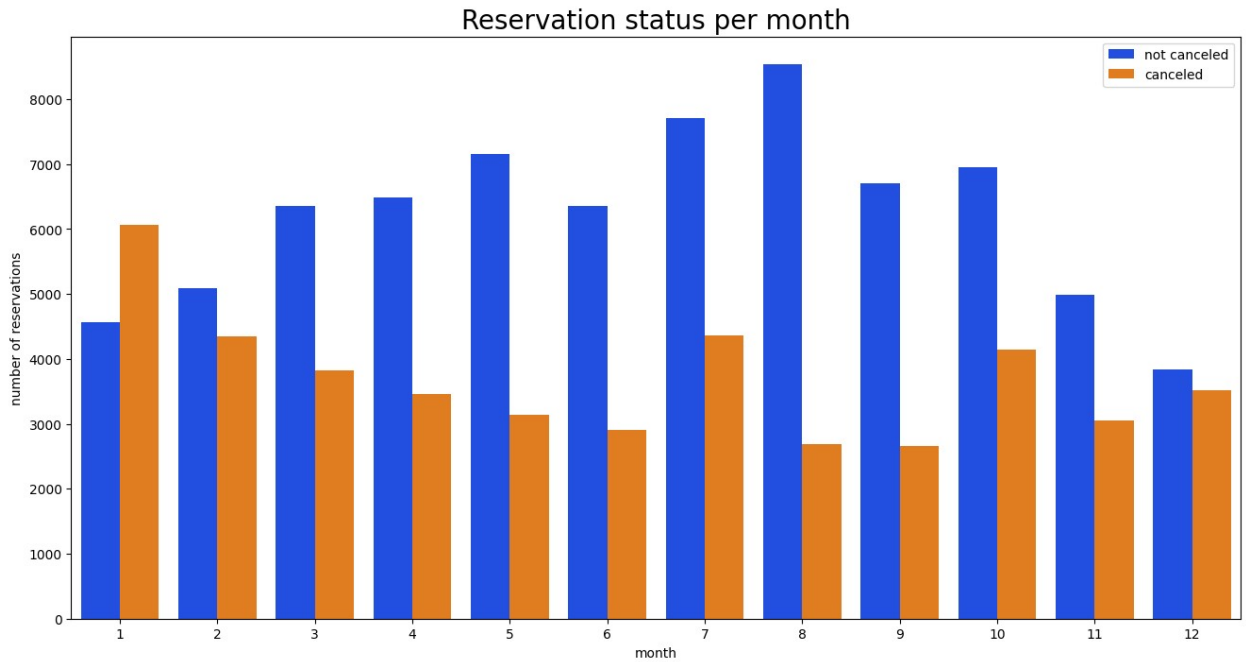
```
resort_hotel = var[var['hotel'] == 'Resort Hotel']
resort_hotel['is_canceled'].value_counts(normalize = True)

is_canceled
0    0.72025
1    0.27975
Name: proportion, dtype: float64

city_hotel = var[var['hotel'] == 'City Hotel']
city_hotel['is_canceled'].value_counts(normalize = True)

is_canceled
0    0.582918
1    0.417082
Name: proportion, dtype: float64

resort_hotel = resort_hotel.groupby('reservation_status_date')
['adr'].mean()
city_hotel = city_hotel.groupby('reservation_status_date')
['adr'].mean()

plt.figure(figsize = (20,8))
plt.title("Average Daily Rate in City and Resort Hotel", fontsize =
30)
plt.plot(resort_hotel.index, resort_hotel, label = "Resort Hotel")
plt.plot(city_hotel.index,  city_hotel, label = "City Hotel")
```

```
plt.legend(fontsize = 20)
plt.show()
```



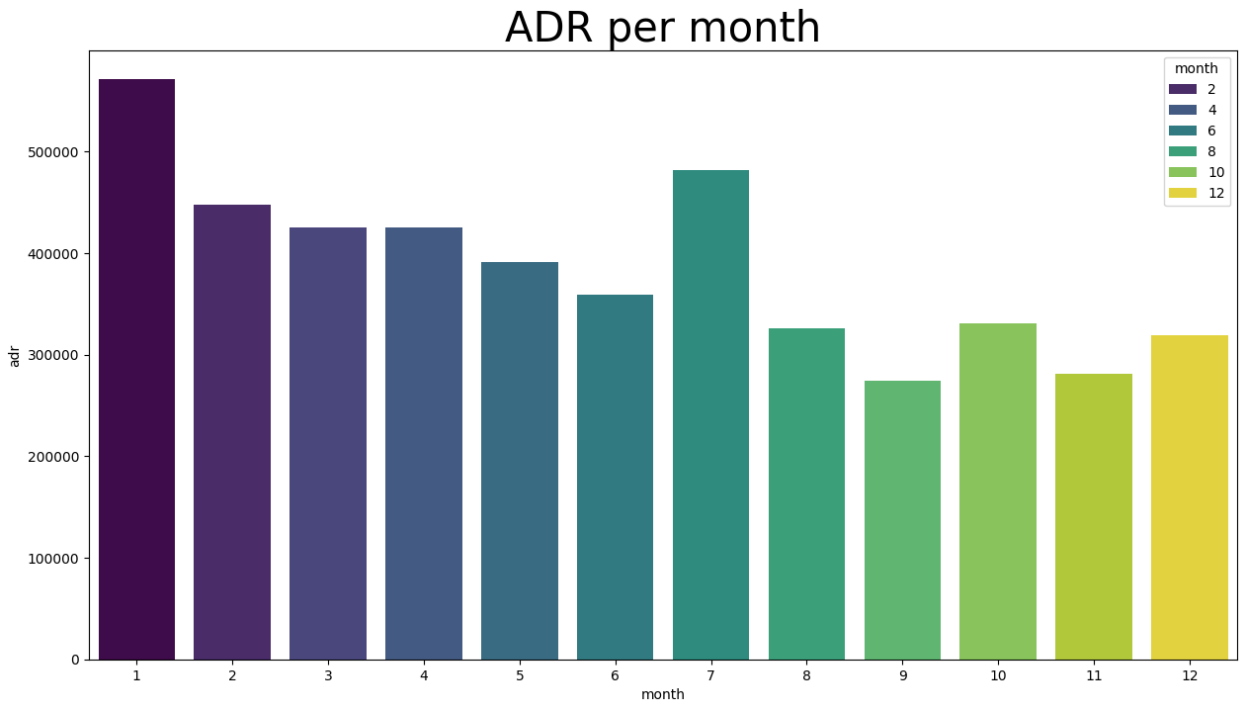
City hotel prices are more stable, while resort hotel prices fluctuate more over time.

```
var['month'] = var["reservation_status_date"].dt.month
plt.figure(figsize = (16,8))
ax1 = sns.countplot(x = "month", hue = "is_canceled", data = var,
palette = "bright")
plt.xlabel("month")
plt.title("Reservation status per month", size = 20)
plt.ylabel("number of reservations")
plt.legend(["not canceled", "canceled"])
plt.show()
```



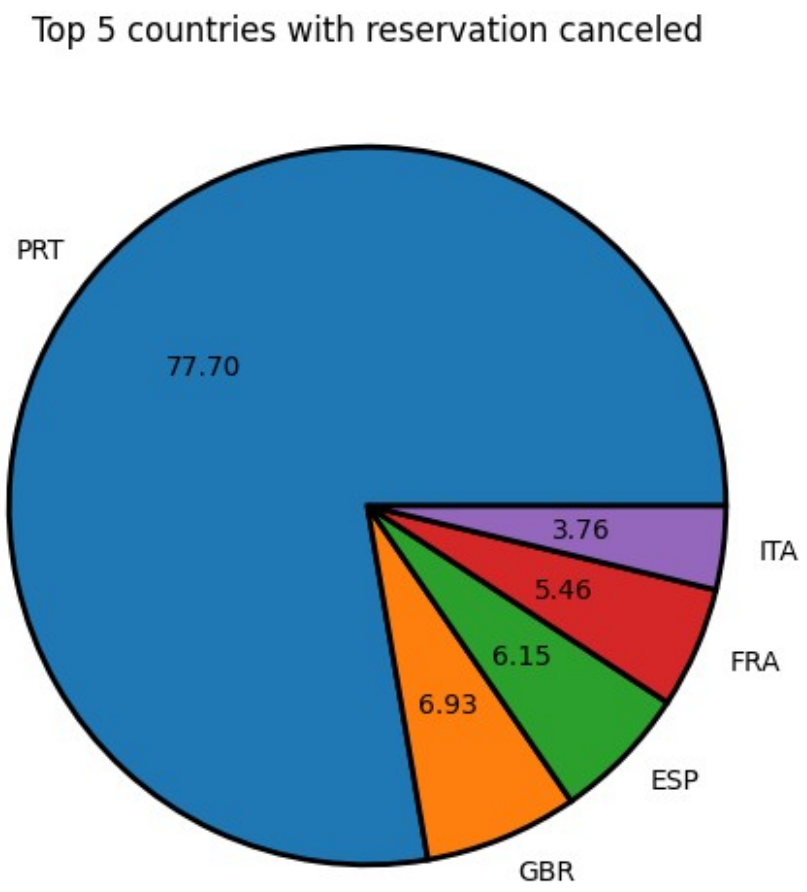
The number of reservations is highest in mid-year months, while cancellations are relatively higher in January and July.

```
cancel_book = var[var["is_canceled"] == 1]
monthly_adr = cancel_book.groupby('month')['adr'].sum().reset_index()

plt.figure(figsize = (15,8))
plt.title("ADR per month", fontsize = 30)
z = sns.barplot(x = "month", y="adr", data = monthly_adr, hue =
"month", palette = "viridis")

plt.show()
```



ADR is highest in January and July, and lowest around September.

```
cancelled_data = var[var["is_canceled"] == 1]
top_5_country = cancelled_data['country'].value_counts().head(5)
plt.figure(figsize = (6,6))

plt.title("Top 5 countries with reservation canceled")
plt.pie(top_5_country, autopct = "%.2f", labels = top_5_country.index,
       wedgeprops = {
           "linewidth" : 2,
           "edgecolor" : "black"})
plt.show()
```



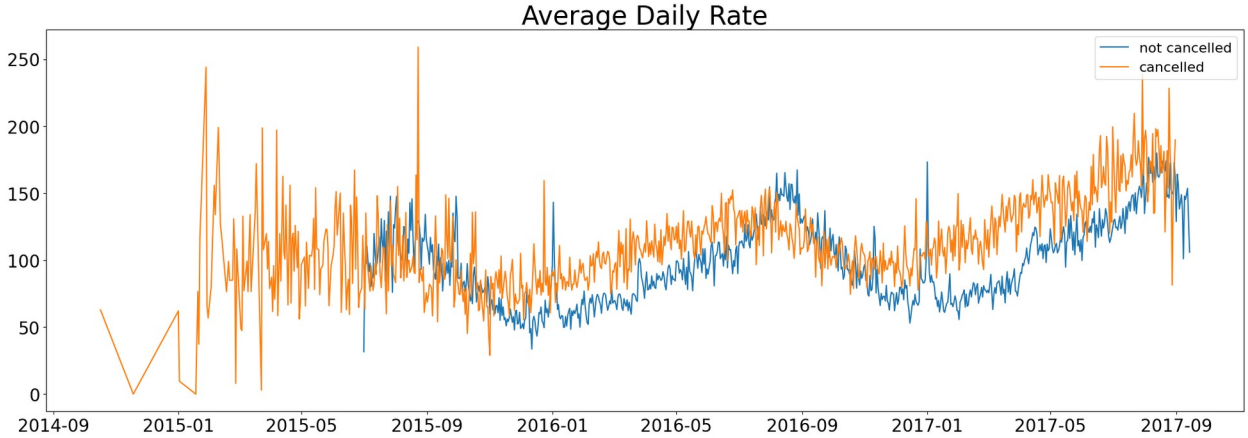
Most cancellations come from Portugal, while other countries contribute much less.

```
var["market_segment"].value_counts()

market_segment
Online TA        56402
Offline TA/TO    24159
Groups           19806
Direct           12448
Corporate         5111
Complementary      734
Aviation           237
Name: count, dtype: int64

var['market_segment'].value_counts(normalize = True)

market_segment
Online TA        0.474377
Offline TA/TO    0.203193
Groups           0.166581
Direct           0.104696
```

```
Corporate        0.042987
Complementary    0.006173
Aviation         0.001993
Name: proportion, dtype: float64
```

cancelled_data["market_segment"].value_counts(normalize = True)

```
cancelled_data = var[var["is_canceled"] == 1]
cancelled_adr = cancelled_data.groupby("reservation_status_date")
["adr"].mean().reset_index()

not_cancelled_data = var[var["is_canceled"] == 0]
not_cancelled_adr =
not_cancelled_data.groupby("reservation_status_date")
["adr"].mean().reset_index()


plt.figure(figsize = (25,8))

plt.title("Average Daily Rate", fontsize = 30)
plt.plot(not_cancelled_adr["reservation_status_date"],
not_cancelled_adr["adr"], label = "not cancelled")
plt.plot(cancelled_adr["reservation_status_date"],
cancelled_adr["adr"], label = "cancelled")
plt.legend(fontsize = 16, loc = "upper right")
plt.xticks(fontsize = 20)
plt.yticks(fontsize = 20)
plt.show()
```



Higher price bookings are more likely to be cancelled.